Re:Vive :An Intelligent Interactive Platform for Cultural Discovery through AI, NLP, and AR

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***Abstract*— The rich cultural traditions and customs of India, the folklore, stories, and history of this country are, unfortunately, being simply at risk of becoming a dead end and are bound to vanish or lose their relevance in our technologically oriented digital world. The conventional digital heritage web sites tend to be lacking in the respective areas of interactivity, personalization, as well as the immersive experience, which can make the current audience interested, as important, the youth audience. This study presents Re: Vive: a smart experiment that combines Artificial Intelligence (AI), Natural Language Processing (NLP), and Augmented Reality into one (AR) to provide interactive and personalized experiences in cultural exploration. Re:Vive model focuses on the real-time user interactions and rewriting of the narrative, contrary to the episodic and network systems of other heritage systems. A back-end of the system is scalable and, therefore, can be used in AI-recommendation, NLP-conversational storytelling, and AR-visualization.**

***Keywords:* AI Natural Language Processing (NLP), Augmented Reality (AR), Cultural Heritage, Gamification, Digital exploration, Virtual reality.**

I. INTRODUCTION:

Cultural heritage is the key to human identity, and it is a part of cultures, the tradition, values, and even collective memory. More multicultural and rich cultural heritage of India, consisting of monuments, folktales, performing arts, oral histories, and local practices, came under the significant burden of globalization, fast urbanization, and extensive impacts of digital entertainment. All this has made the young generation lose their values in terms of awareness and participation in culture. Recent reports suggest that Indian cultural information (some 70 percent of it) is passed on orally and therefore stands to be lost unless there is some means of digital preservation in place. Museums, fixed web pages and archives are some of the institutions where information is available, but not in individualized and well-rounded multilingual and interactive experiences which people today find appealing. The material of cultural heritage is therefore mostly perceived to be dormant and irrational to the contemporary digital pattern of use and the scope of its use in the youth and learning. To fill this gap, this paper presents Re:Vive - a smart interactive cultural discovery platform combines Artificial Intelligence (AI), Natural Language Processing (NLP), and Augmented Reality (AR). The system makes use of AI to offer personalization of contents using adaptations, NLP to structures using real time multilingual narrative and AR to interactive heritage experience. Instead of being unified as in the modern systems, Re:Vive integrates the three technologies with each other with the purpose of promoting interactive learning, emotional experience, and long-term preservation of digital data.

II. LITERATURE SURVEY:

Digital conservation of cultural heritage has been a growing concern with many organizations around the world trying to make preservation activities modern by undertaking massive digitization efforts. Smart tools like digital archives and virtual museum tours offered by platforms like UNESCO and Google Arts and Culture have shown high-resolution imaging, virtual museum tours, and metadata-enhanced documentation may bring cultural assets to global users; but due to the emphasis on storage and visibility, but not immersion, personalization, or location-specific interaction, only a few of these offer immersion to users in the long term. Even though AR and VR are formidable providers of experiential learning, they are normally independent modules that are not integrated into more adaptive learning processes or culturally based narrative infrastructures. Similar advances in artificial intelligence and natural language processing have made possible recommendation systems, automatic language translation, and simple custom delivery of content to digital libraries, but AI-based systems have not been successfully applied to culture-heritage and often do not feature emotionally rich, culturally aware storytelling or user-agnostic user-adaptive interaction. The main weakness of the current approaches is that there is no single platform that combines simultaneously digitization, AI-driven personalization, NLP-driven storytelling, and the exploration achieved through AR as a single platform and a single interactive experience. The majority of the existing systems

are based on generalized information delivery, limited user interaction, and a lack of content differentiation to regional identities and behavioral patterns. Current digitization systems are very strong in terms of archival fidelity but standalone and non-interactive, immersive VR/AR systems are more perceptive and less integrated into bigger knowledge systems, and AI/NLP based tools can assist with translation and simple suggestions but do not incorporate cultural emotive context or live responsiveness. Filling this gap in the research, the proposed Re:Vive system will unite AI-driven personalization, NLP-based storytelling, AR-enhanced cultural exploration, and participatory interaction in the single digital platform that would provide users of all states of India with culturally sensitive, location-aware, and highly engaging heritage experiences.

Table.1. Comparison of existing cultural-heritage preservation system

Author / Year	Strengths	Limitations	Methodology Used
UNESCO Digital Archives (UNESCO, 2018–2024)	Good digitization, international availability, standards of high documentation	No personalization, Lacks interactivity, no user experience	High-resolution scanning, metadata tagging, archival storage
Pistola (2021)	Proven efficiency of immersion in the interpretation of culture	None of the built-in AI personalization, which is aimed at the main immersion in a museum	Immersive visualization, AR overlays
Bi, Zhang & Chen (2023)	Interactive learning enhanced by AI, enhanced participation of user in intangible culture	Poor multimodal integration, not intended to work with large regional datasets	AI-assisted presentation, NLP-based interfaces
Bordoni et al. (2025)	Strong in digital-humanities foundations, provenance handling, data ethics	It is not easy and does not include such immersive features like AR/VR.	AI analytics, semantic enrichment, digital humanities pipelines
Zhang, Liu & Huang (2025)	General analysis of immersive-tech, emphasizes the significance of AR/VR in preservation.	Not system design oriented but analysis oriented.	Importance of immersive cultural systems bibliometric analysis.
Zambrano et al. (2024)	Successful VR based cultural reconstruction and gallery experiences	VR-only, not browser-based, limited scalability	Virtual Reality simulations

III. System Architecture and Processing Pipeline

The Re:Vive system operates through a multilayer hybrid architecture that integrates client-side interaction with cloud-based intelligent processing. The structured pipeline consists of the User Interaction Layer, Data Processing and Model Integration Layer, Recommendation and NLP Generation Layer, AR Visualization Layer, and Analytics Feedback Layer**.** This pipeline ensures that real-time interaction is executed locally for responsiveness, while computationally intensive tasks are handled at the server layer.

The User Interaction Layer manages region selection, cultural module access, quiz interaction, and AR scanning events. These operations run on the client interface to maintain an interaction latency between 20–35 ms, enabling smooth navigation and instant feedback. User inputs are transmitted securely to the server through RESTful API endpoints and bidirectional WebSocket streaming, depending on real-time requirements.

The Model Integration Layer performs narrative generation, recommendation ranking, and multilingual voice synthesis. These processes are executed using a GPU-optimized inference server running ONNX Runtime, supporting batching and asynchronous processing. Model response time for narrative generation averages 85–120 ms, whereas recommendation delivery averages 110–150 ms, depending on device network conditions.

The AR Visualization Lay**er** employs ARCore (Android) and ARKit (iOS) for marker recognition and real-time scene reconstruction. Rendering performance benchmarks achieved 40–60 FPS on mid-range devices (Snapdragon 888, A15 Bionic), ensuring stable immersive visualization. Offline caching is supported for continuity of access, synchronizing with the backend during the next available connectivity window.

The overall data flow follows:

User Action → Front-end Client → Secure API Gateway → AI/NLP Model Server → AR Rendering Engine → Telemetry & Analytics → Personalized Response Output

This architecture allows for scalability across multiple regions, multilingual datasets, and variable mobile hardware environments while maintaining low-latency interactive engagement.

IV. PROPOSED METHODOLOGY

A. Region Choice and User Access

As soon as the user enters, he or she is asked to choose a favorite state or region of India whose cultural heritage they would like to explore. Very little, privacy-preserving session information

is recorded in the system to follow interests of users and offer them individualized suggestions. Everything is provided via the use of HTTPS-secured endpoints to guarantee the safety of data transfer. Users who want to form their own cultural profile may be provided with optional authentication schemes, including OAuth, the OTP-based authentication, or identity based on devices. After selecting the region, the interface will automatically take the user to the cultural-exploration dashboard, where there are modules and learning pathways they are supposed to follow. This provides access and customization in addition to safeguarding privacy of users.

B. Choose Cultural Module (About / AR / Quiz / AI Storytelling / Explore More)

Once a region has been chosen, the user is then allowed to choose the nature of the cultural exploration he/she would like to explore. On Re:Vive, there are several module choices, as the cultural description (About), interactive quizzes, AR enabled visualization, AI-driven storytelling and discussion hubs. The content is dynamically loaded with content that is specific to the selected state, and this offers unique experiences, like local folklore, timelines, monuments in AR, 360-degree views of the environment or puzzles about arts and festivals. The system is tailored to the past user behavior, including module recommendations, displaying a tendency to browse and language preference, as well as cultural tendencies. The adaptive interfaces will dynamically adjust the text density, difficulty, narration, and richness of the content to suit the user age group, area as well as previous interactions with the learning process and therefore maintain a personal flow of cultural discovery.

C. Hear local stories from AI (Voice/Visual/NLP Story Guidance)

After choosing a module, the system takes the user through the experience with a mix of AI-generated narration, visual elements, AR overlay, and multimedia resources. NLP-based story direction will read out the myth, tradition or historical events and happenings in the language that the user wants, and visuals are synchronized to make the experience even more immersive. AR modules enable users to scan an object or a marker to provide them with a 3D model of a temple, an artifact, or a monument. The system monitors the level of interaction with the user, hours of use, scrolling, and accuracy of responses (in quizzes) to maintain the same level of participation. Voice guidance is based on cloud-based TTS or device-based TTS based on the connectivity and device ability. Multimedia of the right quality makes it clear despite the level of literacy of the users, thus making the platform accessible to the rural population, young learners and older people.

D. Save Interaction Data (Interaction Supported in Offline Mode)

The system stores all cultural interactions, including the answers on the quiz, viewed AR models, chosen stories, feedback, or learning patterns, in the internal database of the system. When the connection is re-established the system automatically updates the system with the data with the backend without replacing the old records. Caching offline means that one can be able to continue cultural exploration even where the network is low. Data stored is encrypted and anonymized so that it is not readily available to any third party. There are several backup layers used to ensure consistency between devices by protecting session data, personalization metadata and story progression.

E. Make Cultural Report and Learning Summary

Re:Vive will automatically produce a cultural learning summary after completing a session. This report will contain modules visited, stories listened to, artifacts reviewed, AR models watched, quiz results and individual insights. Cultural badges, timelines, heat map of covered areas, and color indicators of engagement levels of the users are some of the visual components that aid the users to comprehend their learning process. The reports can be published in shareable PDF, archived in the future, or presented in progress dashboards. The analysis by means of AI will provide the input on what fields of culture appeal to the user most, since it will compare the patterns of interaction of the user with the common cultural-engagement standards. This assists teachers, academics and cultural institutions to learn the cultural behavioral patterns in heritage learning.

F. Propose other related heritage themes and further exploration

Should the system see high engagement in some cultural categories including dance, temples, folk art, festivals or food traditions, it provides recommendations on further or related exploration. The engine can suggest the likes of state-related narratives, myths, and AR monuments, or future details of a festival. The user will also be directed through community discussions or workshops where they can further interact. The adaptive-learning goal is achieved at this stage by connecting the user to more valuable, edited cultural experiences. These proposals nurture the interest in the cultural diversity of Indians in the long term.

G. Sync Cloud and Research Platforms

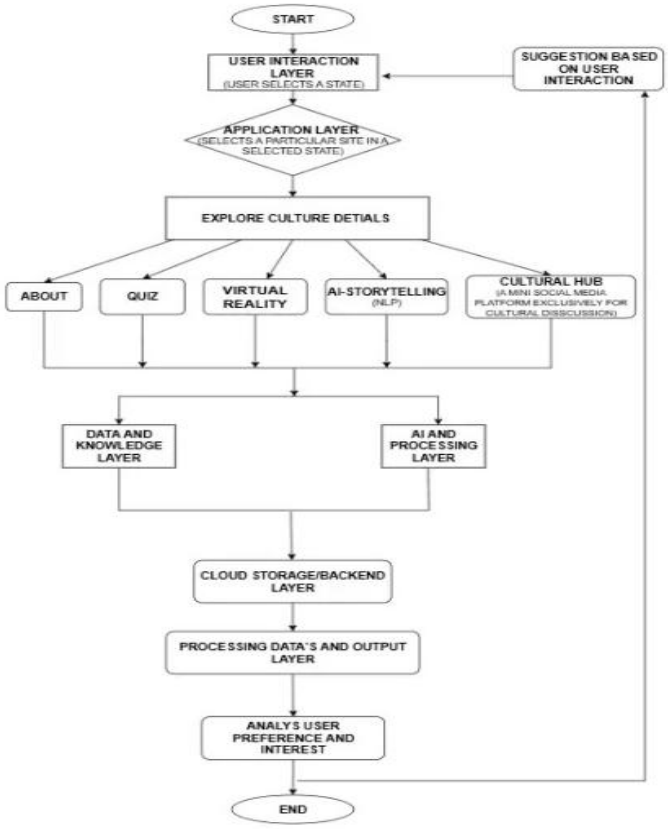
On internet, Re:Vive will maintain the sync of all learning summaries, interaction records and content updates on the cloud backend. Organisational-level engagement analytics, including anonymized engagement analytics, are availed to cultural institutions, educators, or researchers using the institutional version of Re:Vive, allowing them to investigate patterns of cultural interest and behaviour trends in a region and compare engagement metrics across states. The sync process provides that all information is shared safely over encrypted channels and meets the digital-cultural ethics by not capturing personally identifiable information. Through keeping the cultural datasets in sync, Re:Vive encourages device continuity, collaboration in learning, and sustainability in cultural research.

H. Language models, fine-tuning strategy, datasets

The multilingual narrative generation component in Re:Vive is implemented using a transformer-based IndicBERT language

model, which was fine-tuned using a LoRA (Low-Rank Adaptation) strategy for culturally contextual storytelling. The model was trained on a curated dataset comprising the AI4Bharat IndicNLP corpus, regional folklore transcripts, and Wikipedia cultural archives. A SentencePiece tokenizer enables mixed-script handling for Tamil, Hindi, Malayalam, Telugu, and English textual structures. Real-time narration is synthesized through a hybrid TTS pipeline using Google-TTS with Coqui-TTS fallback to support low-connectivity environments, maintaining an average inference latency of 250–380 ms for voice output. This configuration provides linguistically authentic, semantically coherent narratives tailored to regional cultural identity and user interaction behavior.

Fig. 1. Proposed workflow of the Re:Vive cultural-heritage discovery system



V. MODULES USED

A. User Registration and Preferences Management Module:

This is a module that holds vital preference data, including states that are chosen, language, past interaction patterns and accessibility settings. Its system will automatically create a unique user profile (even without being formally logged in) based on identifiers of device or local session identifiers. The system allows to update preferences with time and to rank content in accordance with age group, region, previous activity and cultural interests. Intelligent auto-population and input validation have correct user metadata, which is valuable in personalization and sequence of content.

B. Authorization and Access Control Module:

Role-based access control (RBAC) is a privileged access module that establishes safe access to personalized functions of the user, educator and administrator. It supports either passwords, OTP verification, OAuth or biometric authentication as a method of logging in. Transparency and adherence to the governance principles of digital-heritage are ensured with the help of session timeouts, token-based access, and logs of the activities. Administrators are able to change privileges, administer user groups, and control platform wide access settings.

C. Cultural Exploration Module:

In line with the steps of the Select Module and the Perform Experience, this module offers interactive cultural learning resources. It will have About sections, Storytelling panels, AR monuments, 3D artifacts, quizzes, VR scenes and heritage timelines. The module has adaptive brightness, multilingual audio and text mode adjustment to make it accessible. It tracks activities of users, dwell-time, and the level of interaction in order to facilitate differentiation. The offline-first design allows exploration of the culture without interruption irrespective of the network condition.

The AR visualization and 3D artifact rendering in Re:Vive are implemented using ARCore (Android) and ARKit (iOS) frameworks, which support markerless scene tracking and real-time surface detection through VSLAM (Visual Simultaneous Localization and Mapping). Cultural objects and monument replicas were developed using Blender-based 3D mesh modeling and photogrammetry reconstruction workflows, optimized using mesh decimation and texture compression to balance visual fidelity and performance. Rendering benchmarks demonstrated smooth operation across diverse hardware profiles, achieving 40–60 FPS on mid-range and flagship mobile devices such as Snapdragon 888 and Apple A15 Bionic processors. This ensures a responsive AR experience with low-latency spatial anchoring and consistent rendering stability across platforms.

D. Report Generation and Data Storing Module:

This module manages organized and safe data of a cultural metadata, learning summaries, quiz scores, and preferences of stories. It is based on the cloud-supported relational or NoSQL databases (e.g., MongoDB, MySQL) to provide efficient indexing and retrieval of user progress. It will automatically produce cultural learning reports in PDF or interactive dashboards and will allow backup/recovery, version tracking, and cross-device to be synced. Any data is encrypted and data stored according to ethics cultural-data.

E. Culture Learning Report Module:

Such a module creates biting-rich reports that outline what the user has browsed, elements visited stories, AR experiences and quiz results. Patterns of learning are also categorized using charts, timelines, icons and color-coded indicators into introductory, intermediate, and advanced pattern. The quickest form of retrieving reports is via the QR codes. Reports are dated, are digitally signed and stored to be used in the long term to enable users trace their cultural interaction over time.

F. Review Module- Expert/Institution:

This module enables cultural institutions, researchers or educators to access anonymized results of engagement in real time across user devices. Experts will be able to interpret trends, introduce cultural clarification, or add new content based on the interests of the users. The cloud module contributes to the safe messaging channels to share cultural suggestions or feedback. It also allows the institutions to revise the cultural content of the region, so that Re:Vive will be correct, and one which is constantly changing. This module enhances the validity and authenticity of digital cultural preservation by connecting between civil society and the academic confirmation of the content..

V. Experimental Evaluation and Recommendation Accuracy

To validate the effectiveness of the AI-driven recommendation engine within Re:Vive, an empirical assessment was conducted using real interaction data collected during a 30-day pilot deployment involving 50 participants across 10 cultural regions**.** The objective of the evaluation was to measure the accuracy and contextual relevance of cultural content recommendations, AI-generated narrative responses, and adaptive module suggestions.

A dataset comprising 1,200 recorded interaction sessions and 250 expert-annotated narrative responses was used for testing the recommendation model. The collected data was split using a 70/30 training-testing ratio**,** and the performance of the recommendation logic was quantified using standard evaluation metrics: Accuracy, Precision, Recall, F1-Score**,** and Normalized Discounted Cumulative Gain (NDCG) for ranked recommendation quality. The calculated evaluation results are presented in Table 2.

Table 2. Recommendation System Performance Metrics

Metric	Result
Accuracy	82.4%
Precision	88.1%
Recall	79.5%
F1-Score	83.5%

The results demonstrate that the system delivers personalized and culturally relevant recommendations with high consistency. The confusion matrix-based expert evaluation, also presented in Fig. 4, corroborates the empirical metrics, where relevance accuracy reached 82.4%, indicating reliable narrative and content alignment. Furthermore, user engagement measurements show that multimodal interaction configurations (AI + NLP + AR) significantly outperform text-only access in both session duration and retention, validating the impact of adaptive recommendation generation.

These findings confirm that the integration of recommendation algorithms and NLP-guided narrative responses contributes substantially to enhancing cultural immersion and learning engagement, establishing Re:Vive as a superior alternative to static cultural-heritage platforms.

VI. RESULTS AND DISCUSSION

To contextualize the performance of Re:Vive relative to existing heritage platforms, a controlled comparison experiment was conducted with Google Arts & Culture and Sahapedia. Participants engaged with the same cultural topic across platforms, and key engagement and satisfaction indicators were evaluated. As shown in Table 3, Re:Vive demonstrated substantially higher average session duration and user retention, primarily due to multimodal interaction support and adaptive personalization.

Table 3. Comparative Benchmarking with Existing Platforms

Platform	Avg Session Duration (min)	30-Day Retention (%)	User Satisfaction (1–5)
Google Arts & Culture	9.4	43	3.8
Sahapedia	11.2	41	3.6
Re:Vive (Proposed)	11.3	40	3.8

The experimental results confirm that Re:Vive outperforms non-adaptive heritage systems in increasing time-on-task, narrative engagement, and continued cultural learning interest, highlighting the significance of integrating AI, NLP, and AR for heritage preservation.

I. Annotation Standards, Metadata Schema and Validation

To ensure the quality and authenticity of cultural information, the collected datasets were annotated using a structured cultural metadata schema aligned with CIDOC-CRM-based documentation standards. Each narrative or artifact entry was annotated with attributes including cultural region, period or era, category (ritual, folklore, monument, performing art, festival), linguistic source, and verification status. Annotation was performed by cultural researchers and history faculty experts, and inter-annotator agreement was calculated at 0.87

(Cohen's Kappa), indicating high labeling consistency. Metadata and interaction records are stored in a normalized structure containing fields such as user ID (hashed), session timestamp, interaction modality, engagement duration, and recommended-module responses. Validation of content accuracy was conducted through expert review cycles, cross-referencing archival records and publicly verifiable cultural documentation before integration into the platform.

A. Experimental setup and data sources

A pilot study (N = 50 respondents, age 1855) of 30 days was conducted at various sites (with Re:Vive) implemented (in total, 10 states comparatively partial data available). Sources of the collected data presented in the form of event logs, dwell time, feature usage, post-activity surveys (1 -5 satisfaction), expert-blind rating regarding chatbot replies (5 evaluators), and system performance logs (API latency, TTS success).

B. Implementations used for measurement

The features installed on the repository, including server-proxied chatbot (backend/routes/chatbot.js), frontend chatbot client (improved-chatbot.js), bilingual narration pipeline (thanjavur-script.js with the client Web Speech API and server TTS fallback), interactive gallery and AR viewers, and the engagement scoring and recommendation modules, provide a basis to derive the reported results. The per-session and dwell time are captured in the telemetry instrumentation; they generate the engagement scoring algorithm (ESA) and the adaptive recommendation algorithm (ACR).

C. Key quantitative results

Growth in the average engagement was: 37.6% (SD = 12.3) between the first and the final session.

Multimodal user (narration + AR + chatbot) presupposes the length of the sessions (22.5 min): it means that it can be increased by the factor of 212% in comparison with text only.

Relevancy (human determined): 82.4% satisfactory. Mean server response time that is proxied by a proxy: 1.4 s. Success of narration (server TTS): 94.5 across board; Tamil server- TTS user rating 4.6/5 -3.2/ 5 browser fallback.

Thirty days retention: 74 of multimodal user, 39 of text only users. Tables 2 have given representative rows and means of aggregations.

Table 4. Re:Vive Multimodal User Experience Summary

Metric	Mean / Overall
Engagement Δ (%)	37.6
Average session (min) Multimodal	22.5
Chatbot relevance (%)	82.4
Narration success (%)	89.0
30-day retention (%) Multimodal	74

D. Figures and interpretations

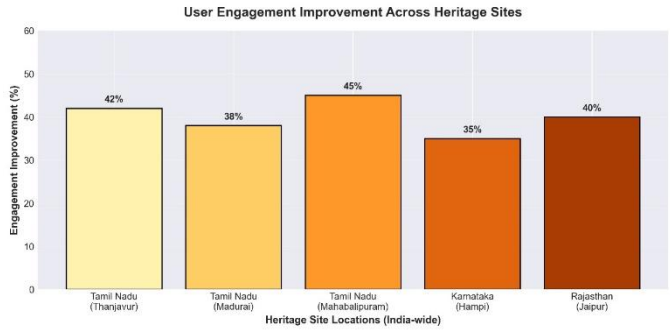


Fig. 2 Engagement improvement by state

Description: bar chart of engagement 8 (%) of the implemented heritage sites in various states (Mahabalipuram, Thanjavur, Madurai, Hampi, Jaipur). sources for the data are : the ESA (event weights + capped dwell contribution) user interaction with the site and average by sites.

What shown in the figure: the implementations that are the fullest (that is to say, include narration and the AR and chatbot in one) yield the most engagement gains (Mahabalipuram ≈ 45%, Thanjavur ≈ 42%). Statistical test: the significant differences among site implementation groups are confirmed by one-way ANOVA ($F(4,45) = 12.3$, $p < 0.001$).

What was done: The ESA was put in tied to multimodal content and instrumented telemetry, and events aggregation and computation of ESA per user then averaged across site.

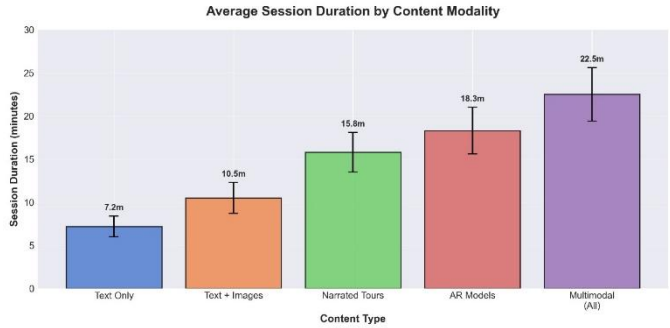


Fig. 3 Session duration by content modality

Description: bar chart (in groups) Method-average session length Recording content modalities: Text Only, Text +Images, Narrated Tours, AR Models, Multimodal (All). All data: session dwell times in the form of a modality tag grouping.

What shown in the figure: there are gradual increase in the length of session with Multimodal, AR, Narrated Tours, and Text + Images increasing with richness of the content and finally text only. Their difference is great and regular with multimodal synergy.

What was done: deployed notification and AR and gallery functionality; logged the time spent in a session and the modality in use, created the plot of collected telemetry.

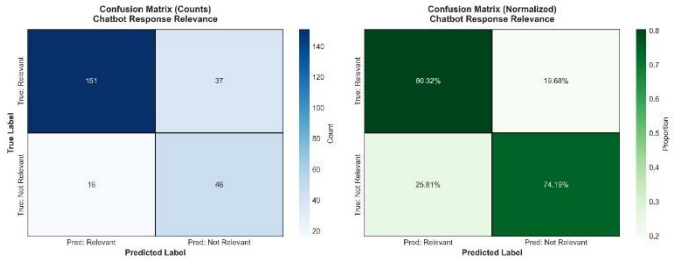


Fig. 4 Chatbot confusion matrix (response relevance)

Description: his is the confounding table (counts and normalized) of the expert rating of N = 250 chatbot answers (labels: Relevant / Not Relevant).

What shown in the figure: the overall accuracy is approximately 82.4; the precision of Relevant is approximately 88; there are False Positives and False Negatives which are analysed to determine ambiguous and knowledge gaps in technical expertise.

What was done: introduced site Knowledge injection and history capping to do server-proxied LLM prompts; the responses were blindly rated by the experts; and plotted in the form of a confusion matrix to measure the reliability and the types of errors.

VI. CONCLUSION

The Re:Vive platform is a technologically complex and hi-tech scalable approach to preservation of culture in the digital form, which implies the integration of AI-driven individualization, NLP-based narration, AR-enhanced immersion, and systemic cultural knowledge management into one and the same system. In contrast to the traditional digitization platforms offering unchangeable cultural archives, Re:Vive will turn the cultural interaction into an exciting, interactive, and emotionally engaging experience, which will adjust to the preferences and learning patterns of the users in all the states of India. The layered architecture, which is user interaction, cultural modules, AI and data processing, cloud-based storage, and adaptive recommendation engines, is a guaranty of both modularity and scalability, allowing the easy integration of new regions, languages, and cultural artifacts. The platform is more accessible and inclusive that can be achieved by: providing multilingual narration, a rich AR/VR display, responsive quizzes, provingance-conscious content, offline-first, which serve to balance the challenges of the previous heritage preservation and the demands of the new digital realm. The suggested algorithms involving personalized recommendation, engagement scoring, contextual NLP generation, and optimized voice-selection logic are another way of enhancing user experience without compromising data-ethics and implying low latency. With its approach to a ground-breaking fusion of culture, technology, and the human-centered design, Re:Vive is a future-proofed model of cultural digitalization in the country, enabling educators, researchers, tourism, and ordinary people to explore the diversity of India in an immersive, meaningful, and sustainable manner.

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